

## »» Chapter XIII ««

# HARMONIC AND SUBHARMONIC FUNCTIONS

### PREFATORY REMARKS

For many purposes of modern complex function theory the theory of harmonic functions and the allied theory of subharmonic functions, which was introduced by F. Riesz, are well-nigh indispensable. They play a useful and natural role in such questions as the study of Hardy classes and the existence theorems of the theory of Riemann surfaces. Because of the limitations imposed on the present book, our appeal to the methods of real analysis is restricted. We shall treat only some of the simplest aspects of the theories of harmonic and subharmonic functions. An introductory account of the theories of harmonic and subharmonic functions is to be found in Chs. IV and V of [53] where the approach and the questions considered differ in many respects from those of this chapter. We are operating here with no appeal to a refined use of real analysis and are pursuing utilitarian goals. Useful references are [78], [102].

We treat the Dirichlet problem for a disk (via the Poisson integral) and thereupon immediate function-theoretic applications. These include the Schwarz reflexion principle and the extension principle of Carleman, which appears as a consequence of it. We introduce subharmonic functions in a restricted setting above all with the specific object that it serve as a step toward the existence theorem of Ch. XVIII wanted for the solution of the problem of Riemann and Klein; to be sure, the results will have to be trans-

planted to Riemann surfaces, but this will involve no difficulties. Of course, quite different applications will be seen to emerge as well.

The exposition of this chapter presupposes the developments of §8, Ch. IV and §3, Ch. XI.

## 1. POISSON INTEGRAL AND APPLICATIONS

Let  $u$  be a continuous real-valued function on  $\overline{\Delta(0; 1)}$  which is harmonic in  $\Delta(0; 1)$ . Since there exists  $f$  analytic on  $\Delta(0; 1)$  such that  $\operatorname{Re} f(z) = u(z)$ ,  $|z| < 1$ , we have by (5.7), Ch. IV, when  $0 < r < 1$ , the formula

$$u(rz) = \frac{1}{2\pi} \int_0^{2\pi} u(re^{i\theta}) k(\theta, z) d\theta, \quad |z| < 1,$$

where

$$k(\theta, z) = \operatorname{Re} \left( \frac{e^{i\theta} + z}{e^{i\theta} - z} \right) = \frac{1 - |z|^2}{|e^{i\theta} - z|^2}, \quad (1.1)$$

and hence, taking the limit at 1, we have

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\theta}) k(\theta, z) d\theta. \quad (1.2)$$

It follows from (1.2) (as well as from the uniqueness theorem given in Ex. 8.5, Ch. IV) that  $u$  is determined by  $u|C(0; 1)$ . That is, if  $v$  also satisfies the conditions stated for  $u$  and  $u(z) = v(z)$ ,  $|z| = 1$ , then  $u = v$ . But (1.2) goes beyond the uniqueness theorem in that it actually specifies the values taken by  $u$  at the points of  $\Delta(0; 1)$  in terms of  $u|C(0; 1)$ . We now ask whether there exists  $u$  satisfying the stated conditions and

$$u|C(0; 1) = U, \quad (1.3)$$

where  $U$  is a given continuous real-valued function on  $C(0; 1)$ . Our question is just the Dirichlet problem for the unit disk for the case of an assigned continuous real-valued boundary function. The only possibility for a solution is the function  $u$  with domain  $\overline{\Delta(0; 1)}$  satisfying (1.3) and

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} U(e^{i\theta}) k(\theta, z) d\theta, \quad |z| < 1. \quad (1.4)$$

The right side is termed *the Poisson integral formula associated with  $U$*  and it defines a function that is the real part of a function analytic on  $\Delta(0; 1)$  by (1.1) and hence is harmonic on  $\Delta(0; 1)$ .

We shall be assured that  $u$  specified by (1.3) and (1.4) is the desired solution

of the stated problem once we show that it is continuous at each point of  $C(0; 1)$ . Applying (1.2) to the constant 1 we see that

$$\frac{1}{2\pi} \int_0^{2\pi} k(\theta, z) d\theta = 1, \quad |z| < 1.$$

Suppose that  $\alpha \in \mathbf{R}$ ,  $0 < \delta < \pi$ . We consider

$$\int_{\alpha+\delta}^{\alpha+\pi} k(\theta, z) d\theta + \int_{\alpha-\pi}^{\alpha-\delta} k(\theta, z) d\theta \quad (1.5)$$

(with domain  $\Delta(0; 1)$ ) and seek its limit at  $e^{i\alpha}$ . From (1.1) we obtain

$$k(\theta, z) \leq \frac{1 - |z|^2}{m^2}$$

when  $\delta \leq |\theta - \alpha| \leq \pi$ ,  $|z| < 1$ ,  $|z - e^{i\alpha}| \leq m$ . Here

$$2m = \min\{|e^{i\theta} - e^{i\alpha}| : \delta \leq |\theta - \alpha| \leq \pi\}.$$

Consequently the limit of (1.5) at  $e^{i\alpha}$  is 0. We note that for  $|z| < 1$  we may represent  $u(z) - U(e^{i\alpha})$  by

$$\frac{1}{2\pi} \int_{\alpha-\delta}^{\alpha+\delta} [U(e^{i\theta}) - U(e^{i\alpha})] k(\theta, z) d\theta + \frac{1}{2\pi} \int_{\alpha+\delta}^{\alpha-\delta+2\pi} [U(e^{i\theta}) - U(e^{i\alpha})] k(\theta, z) d\theta.$$

The second integral is in absolute value no greater than the value given by display (1.5) times twice  $\max_{C(0;1)} |U|$  and consequently it has limit 0 at  $e^{i\alpha}$ . The first integral is in absolute value no greater than

$$M(\delta) = \max\{|U(e^{i\theta}) - U(e^{i\alpha})| : |\theta - \alpha| \leq \delta\}.$$

Given  $\varepsilon$  positive, we fix  $\delta$  so that  $M(\delta) < \varepsilon$  and conclude that

$$|u(z) - U(e^{i\alpha})| < \varepsilon$$

for  $z$  sufficiently near  $e^{i\alpha}$ ,  $|z| < 1$ . The continuity of  $u$  at  $e^{i\alpha}$  is assured and hence  $u$  is the desired solution of the Dirichlet problem stated above.

We note that if  $v$  is continuous and real-valued on  $\overline{\Delta(a; r)}$  and harmonic in  $\Delta(a; r)$  where  $a \in \mathbf{C}$  and  $r$  is positive, then  $v(a + rz)$  is continuous on  $\overline{\Delta(0; 1)}$  and is harmonic in  $\Delta(0; 1)$ . Hence we conclude that

$$v(z) = \frac{1}{2\pi} \int_0^{2\pi} v(a + re^{i\theta}) k\left(\theta, \frac{z-a}{r}\right) d\theta, \quad |z-a| < r. \quad (1.6)$$

It is also seen that the Dirichlet problem for  $\Delta(a; r)$  with assigned continuous boundary function may be treated with the aid of the Poisson integral formula. Indeed, if  $V$  is a continuous real-valued function on  $C(a; r)$ , then the

solution of the Dirichlet problem with boundary function  $V$  is the function  $v$  on  $\Delta(a; r)$  satisfying  $v|C(a; r) = V$  and (1.6) with “ $v(a + re^{i\theta})$ ” replaced by “ $V(a + re^{i\theta})$ .”

## EXERCISES

1.1. Show that

$$v_z(z) = \frac{1}{2\pi} \int_0^{2\pi} v(a + re^{i\theta}) \frac{re^{i\theta}}{(a + re^{i\theta} - z)^2} d\theta, \quad |z - a| < r.$$

1.2. Let  $O$  be an open subset of  $\mathbb{C}$ . Let  $(u_n)_{n \in \mathbb{N}}$  be a sequence of functions harmonic on  $O$  which tends uniformly to  $u$  in  $O$  (in the sense of  $\mathbb{R}$ ). Show with the aid of (1.6) that  $u$  is harmonic on  $O$ .

1.3. Show that

$$\int_{-\infty}^{+\infty} \operatorname{Im} \frac{1}{t - z} dt = \pi, \quad \operatorname{Im} z > 0,$$

and that if  $a \in \mathbb{R}$  and  $b$  is positive, the limit at  $a$  of

$$\int_{-\infty}^{a-b} \operatorname{Im} \frac{1}{t - z} dt + \int_{a+b}^{+\infty} \operatorname{Im} \frac{1}{t - z} dt$$

is 0, and develop the properties of the Poisson integral for the upper half plane, namely

$$\frac{1}{\pi} \int_{-\infty}^{+\infty} f(t) \operatorname{Im} \frac{1}{t - z} dt, \quad \operatorname{Im} z > 0,$$

where  $f$  is a continuous real-valued function on  $\mathbb{R}$  subject to a condition of growth to be specified. [For example, everything proceeds well when it is assumed that  $f$  is bounded. Are weaker conditions available?]

1.4. Show with the aid of (1.6) that if  $v(z) \geq 0$ ,  $|z - a| \leq r$ , then

$$v(a) \frac{r - |z - a|}{r + |z - a|} \leq v(z) \leq v(a) \frac{r + |z - a|}{r - |z - a|}, \quad |z - a| < r \quad (1.7)$$

(Harnack inequality). Hint: Use

$$\frac{1 - |z|}{1 + |z|} \leq k(\theta, z) \leq \frac{1 + |z|}{1 - |z|}, \quad |z| < 1.$$

1.5. Demonstrate with the aid of the Harnack inequalities the following theorem: Let  $A$  be a region in  $\mathbb{C}$  and let  $a \in A$ . Let  $\mathfrak{P}$  denote the set of functions  $u$  harmonic on  $A$  which satisfy:  $u(a) = 1$ ,  $u(z) > 0$  for  $z \in A$ . For each  $z \in A$  let  $\lambda(z) = \inf \{u(z) : u \in \mathfrak{P}\}$  and let  $\mu(z) = \sup \{u(z) : u \in \mathfrak{P}\}$ . Then  $0 < \lambda(z) \leq 1 \leq \mu(z) < +\infty$ ,  $z \in A$ , and further  $\lambda$  and  $\mu$  are continuous on  $A$ .

Hint: Consider the sets  $\{\lambda(z) > 0\}$  and  $\{\lambda(z) = 0\}$  and show with the aid of (1.7) that each is open. Proceed similarly with  $\mu$  and thereupon treat the continuity.

Show with the aid of this theorem that if  $(u_n)_{n \in \mathbb{N}}$  is a sequence of functions harmonic on  $A$  satisfying:  $u_n(z) \leq u_{n+1}(z)$ ,  $z \in A$ ,  $n \in \mathbb{N}$ , then either  $(u_n)$  tends uniformly to the constant  $+\infty$  in  $A$  or else  $(u_n)$  tends uniformly to a harmonic function in  $A$  (Harnack convergence principle). Hint: Consider  $(u_n(a))$  for a given point  $a \in A$  and control  $u_{n+m}(z) - u_n(z)$  in terms of  $u_{n+m}(a) - u_n(a)$ ,  $m$  a positive integer.

1.6. Let  $\Phi$  be the class of  $f$  analytic on  $\Delta(0; 1)$ , univalent, and satisfying  $f(0) = 0$  and

$$\inf\{|w|: f(z) \neq w, |z| < 1\} = 1.$$

Given  $f \in \Phi$ , let  $-\bar{\eta} \in C(0; 1)$  be omitted by  $f$  and let  $g$  denote the analytic square root of  $1 + \eta f$  satisfying  $g(0) = 1$ , so that  $g$  composed with the restriction to  $\Delta(0; 1)$  of the inverse of  $\eta f$  is the analytic square root  $S$  of  $1 + z$  (with domain  $\Delta(0; 1)$ ) satisfying  $S(0) = 1$ . Show that

$$|z| \leq |S(z) - 1|(|S(z)| + 1) \leq |S(z) - 1|(1 + \sqrt{2}),$$

$|z| < 1$ . Conclude that the image of  $S$  contains  $\Delta(1; \sqrt{2} - 1)$  and thereupon, invoking the fact that the image of  $g$  contains the image of  $S$  and the univalence of  $f$ , that  $\Delta(-1; \sqrt{2} - 1)$  is in the complement of the image of  $g$ . Thereupon show that there exist positive numbers  $A$  and  $B$  such that

$$|f(z)| \leq A \exp\left(B \frac{1 + |z|}{1 - |z|}\right), \quad (1.8)$$

$|z| < 1$ ,  $f \in \Phi$ , by applying the Harnack inequality to  $\log |(1 + \sqrt{2})(g + 1)|$  and that  $|f(z)| \geq |z|$ ,  $|z| < 1$ ,  $f \in \Phi$ , by applying the maximum principle to  $|z/f(z)|$ . [Actually, a sharp inequality replacing (1.8) may be obtained with the aid of the celebrated  $\frac{1}{2}$ -theorem of Koebe-Bieberbach, cf. [52].]

1.7. Let  $f$  be analytic on a region  $\Omega$  of  $\mathbb{C}$ , where  $0 \in \Omega \subset \mathbb{C}$  but  $\Omega \neq \mathbb{C}$ , which is univalent and satisfies  $f(\Omega) = \Omega$ ,  $f(0) = 0$ . Show that  $|f'(0)| = 1$  (Radó). Hint: Take  $r$  positive so that  $\Delta(0; r) \subset \Omega$  and bring the inequality (1.8) to bear on  $f_n(rz)$ ,  $|z| < 1$ , where  $f_n$  is the  $n$ th iterate of  $f$ , to conclude  $|f'(0)| \leq 1$ .

1.8. Show with the aid of Exs. 1.1 and 1.5 that if  $u \in \mathfrak{P}$ , then

$$|u_z(z)| \leq \frac{\mu(z)}{d(z)}$$

where

$$d(z) = \inf\{|w - z|: w \in \mathbb{C} - A\}.$$

Conclude that if  $A = \mathbb{C}$ , then  $u$  is constant. [Here  $\mu(z)/(+\infty)$  is construed as 0.]

1.9. Selection principle. Let  $(a_{mn}): \mathbb{N}^2 \rightarrow \mathbb{R}$ . We suppose that

$$\sup\{|a_{mn}|: m \in \mathbb{N}\} < +\infty$$

for each  $n \in \mathbb{N}$ . Then there exists  $\mu: \mathbb{N} \rightarrow \mathbb{N}$  increasing such that for each  $n$  the sequence  $(a_{\mu(m), n})_{m \in \mathbb{N}}$  possesses a limit. [It is possible to give many variations of the hypothesis. For our immediate purposes the stated hypothesis of real entries and bounded "columns" is adequate.] Hint: Use Ex. 6.2, Ch. I, and proceed recursively. Let  $b_n$  denote the sequence  $(a_{mn})_{m \in \mathbb{N}}$ . Let  $\nu_0: \mathbb{N} \rightarrow \mathbb{N}$  be increasing and such that  $b_0 \circ \nu_0$  possesses a limit, let  $\nu_1: \mathbb{N} \rightarrow \mathbb{N}$  be increasing and such that  $(b_1 \circ \nu_0) \circ \nu_1$

possesses a limit, and in general let  $\nu_{k+1}: \mathbf{N} \rightarrow \mathbf{N}$  be increasing and such that

$$[b_{k+1} \circ (\nu_0 \circ \cdots \circ \nu_k)] \circ \nu_{k+1}$$

possesses a limit,  $k \in \mathbf{N}$ . Define  $\mu(k) = \nu_0 \circ \cdots \circ \nu_k(k)$ ,  $k \in \mathbf{N}$ , and verify that  $\mu$  serves.

**1.10.** Suppose now that  $A$  is a region  $\subset \mathbf{C}$ ,  $a \in A$ , and  $(u_m)_{m \in \mathbf{N}}$  is a sequence of functions harmonic on  $A$  satisfying  $u_m(z) \geq 0$ ,  $z \in A$ ,  $m \in \mathbf{N}$ , and such that  $(u_m(a))$  is bounded. Show that there exists a subsequence of  $(u_m)$  which tends to a harmonic function uniformly in  $A$ .

Hint: Let  $(z_n): \mathbf{N} \rightarrow A$  be univalent and such that  $\{z_n: n \in \mathbf{N}\}$  is dense in  $A$ . [ $B$  is dense in  $A$  provided that  $A \subset \overline{B}$ .] Let  $\mu$  be such that for each  $n \in \mathbf{N}$  the sequence  $(u_{\mu(m)}(z_n))_{m \in \mathbf{N}}$  possesses a limit. Suppose for notational simplicity that  $(u_m)$  itself enjoys this property (using  $(u_{\mu(m)})$  in place of  $(u_m)$ ). Let  $\overline{\Delta(b; r)} \subset A$ . By Ex. 1.8 there exists  $M$  positive such that

$$|u_m(z) - u_m(w)| \leq M |z - w|,$$

$z, w \in \overline{\Delta(b; r)}$ ,  $m \in \mathbf{N}$ . Note that

$$\begin{aligned} |u_p(z) - u_q(z)| &\leq |u_p(w) - u_q(w)| + |u_p(z) - u_p(w)| + |u_q(w) - u_q(z)| \\ &\leq |u_p(w) - u_q(w)| + 2M |z - w|. \end{aligned}$$

Conclude that given  $\varepsilon$  positive, on taking a finite subset  $W$  of  $\{z_n: n \in \mathbf{N}\} \cap \overline{\Delta(b; r)}$  so that each point of  $\overline{\Delta(b; r)}$  is at a distance less than  $\varepsilon$  from some point of  $W$ , we have (with the aid of the second inequality)

$$|u_p(z) - u_q(z)| \leq (2M + 1)\varepsilon,$$

$z \in \overline{\Delta(b; r)}$ ,  $p, q \geq s$  for some  $s \in \mathbf{N}$ .

**1.11.** Show that if a sequence  $(u_n)$  of functions harmonic on  $O$  open  $\subset \mathbf{C}$  tends uniformly in  $O$  to a harmonic function  $u$ , then  $((u_n)_z)$  tends uniformly in  $O$  to  $u_z$ . Suppose in addition that  $O$  is a region,  $u_n = \operatorname{Re} f_n$  where  $f_n$  is analytic on  $O$ , and  $(f_n(a))$  possesses a limit ( $\in \mathbf{C}$ ) for some  $a \in O$ , and conclude that under these circumstances  $(f_n)$  tends uniformly (in the sense of the  $\mathbf{C}$ -metric) in  $O$  to a function  $f$ .

**1.12.** Let  $O$  be an open nonempty subset of  $\hat{\mathbf{C}}$ . Let  $\Phi$  denote a nonempty family of continuous maps of  $O$  into  $\mathbf{C}$ . The family  $\Phi$  will be said to be *normal* provided that for each  $\varphi: \mathbf{N} \rightarrow \Phi$  there exists  $\nu: \mathbf{N} \rightarrow \mathbf{N}$ ,  $\nu$  increasing, such that  $\varphi \circ \nu$  tends uniformly (in the sense of the  $\mathbf{C}$ -metric) in  $O$  to a function  $f$ . [This definition is slightly more restrictive than that of Montel [85].] Let

$$M(z) = \sup \{|f(z)|: f \in \Phi\}, \quad z \in O.$$

Show that  $M$  is a continuous map of  $O$  into  $\mathbf{R}^+$  when  $\Phi$  is normal.

Hint: Given  $z \in O$  take  $(f_n): \mathbf{N} \rightarrow \Phi$  such that  $\lim_{n \rightarrow \infty} |f_n(z)| = M(z)$  and conclude that  $M(z) < +\infty$ . To show continuity make use of the fact that if  $(f_n): \mathbf{N} \rightarrow \Phi$  tends to  $f$  uniformly in  $O$  and  $(z_n): \mathbf{N} \rightarrow O$  satisfies  $\lim_{n \rightarrow \infty} z_n = z \in O$ , then

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

**1.13.** Let  $A$  be a region  $\subset \mathbf{C}$ . Let  $\Phi$  be a normal family of functions harmonic on  $A$ . Let  $(u_n): \mathbf{N} \rightarrow \Phi$  possess a pointwise limit on a disk (or even on a dense subset of a disk  $\subset A$ ). Show that  $(u_n)$  tends uniformly in  $O$  to a harmonic function.

**1.14.** Let  $A$  be a region  $\subset \hat{\mathbb{C}}$ . Let  $\Phi$  be a normal family of functions analytic on  $A$  and let  $(f_n): \mathbb{N} \rightarrow \Phi$  possess a pointwise limit on  $E \subset A$  where  $E$  clusters at  $a \in A$ . Then  $(f_n)$  tends uniformly in  $A$  to a function  $f$  (theorem of Stieltjes–Vitali).

**1.15.** Let  $O$  be a nonempty open subset of  $\hat{\mathbb{C}}$  and let  $\Phi$  be a nonempty family of functions analytic on  $O$ . Let  $M$  defined as in Ex. 1.12 be bounded on each nonempty compact subset of  $O$ . Then  $\Phi$  is normal (Osgood–Montel). Hint: Selection principle and Ex. 1.10.

**1.16.** Let  $A$  be a region in  $\mathbb{C}$ ,  $0 \in A \neq \mathbb{C}$ . Let  $\Phi$  denote the family of functions analytic on  $A$  which are univalent and satisfy:  $f(0) = 0$ ,  $\min\{|w|: w \in \mathbb{C} - f(A)\} = 1$ . Introduce

$$M_k(z) = \sup\{|f^{(k)}(z)|: f \in \Phi\}.$$

Show that  $M_0$  and  $M_1$  are finite-valued and continuous on  $A$ .

Hint: Consider first the finiteness of  $M_0$  and  $M_1$  applying the standard connectedness argument to  $\text{int}\{M_0(z) + M_1(z) < +\infty\}$ .

**1.17.** Consider the same question where  $A$  is now a region in  $\mathbb{C}$  containing 0 and  $\Phi$  consists of the functions analytic on  $A$  which are univalent and satisfy:  $f(0) = 0$ ,  $f'(0) = 1$ .

**1.18.** Let  $O$  be an open subset of  $\mathbb{C}$ . Let  $u$  be a function real-valued and continuous on  $O$  and suppose that  $u$  enjoys the Gauss mean-value property in the sense that

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta$$

when  $\overline{\Delta(a; r)} \subset O$ ,  $r < +\infty$ . Then  $u$  is harmonic on  $O$  (Koebe). Hint: Consider  $u(z) - v(z)$ ,  $|z - a| < r$ , where  $v$  is the solution of the Dirichlet problem for  $\Delta(a; r)$  with boundary function  $u|_{\partial\Delta(a; r)}$  and note that it enjoys the mean-value property and hence satisfies the maximum principle, and treat  $v(z) - u(z)$  similarly.

**1.19.** Continuing with  $O$  as in Ex. 1.18, let  $f: O \rightarrow \mathbb{C}$  possess a derivative continuous on  $O$ . Show by differentiation under the integral sign but without appeal to the Cauchy theory that  $f$  enjoys the mean-value property and therefrom, using the Poisson integral, that  $f(z)$  admits a representation  $\sum_{k=0}^{\infty} b_k(z - a)^k$ ,  $|z - a| < r$ .

## 2. WEIERSTRASS APPROXIMATION THEOREM AND THE DIRICHLET PROBLEM FOR AN ANNULUS

The Poisson integral formula is an instrument of great versatility. In this section we shall use it to obtain a form of the Weierstrass approximation theorem (2.2) and shall see that we may even allow functions taking values in a complex Banach space. The approximation theorem will then be used to solve the Dirichlet problem for an annulus.

Let  $f$  be a continuous map of  $C(0; 1)$  into a complex Banach space  $V$ . Let

$$u(z) = \int_0^{2\pi} f(e^{i\theta}) \frac{k(\theta, z)}{2\pi} d\theta, \quad |z| < 1. \quad (2.1)$$

We see using a paraphrase of the reasoning of the preceding section that  $f \circ u$  is continuous on  $\overline{\Delta(0; 1)}$ . Hence given  $\varepsilon$  positive there exists  $r$ ,  $0 < r < 1$ , such

that  $\|u(rz) - f(z)\| < \varepsilon/2$ ,  $z \in C(0; 1)$ . On noting that  $u(z) = S_1(z) + S_2(\bar{z})$  where  $S_1$  and  $S_2$  are the sums of power series with coefficients in  $V$  which converge on  $\Delta(0; 1)$ , we see that there exists a sum of the form  $\sum_{-n}^n a_k z^k$ ,  $a_k \in V$ , such that

$$\left\| u(rz) - \sum_{-n}^n a_k z^k \right\| < \varepsilon/2, \quad |z| = 1.$$

Hence

$$\left\| f(z) - \sum_{-n}^n a_k z^k \right\| < \varepsilon, \quad |z| = 1. \quad (2.2)$$

This shows that  $f$  may be approximated uniformly on  $C(0; 1)$  by a sum of the form  $\sum_{-n}^n a_k z^k$ . In the special case of  $f$  real-valued we may assume that  $a_{-k} = \bar{a}_k$ ,  $k = 0, \dots, n$ .

### EXERCISE

**2.1.** Use the approximation theorem just obtained to prove the following Weierstrass approximation theorems. (a) If  $f$  is a continuous periodic map from  $\mathbf{R}$  into  $V$  with period  $2\pi$  and  $\varepsilon$  is positive, then there exists a trigonometric sum  $T(x)$  of the form

$$\sum_0^n (a_k \cos kx + b_k \sin kx), \quad a_k, b_k \in V,$$

such that  $\|f(x) - T(x)\| < \varepsilon$ ,  $x \in \mathbf{R}$ . (b) If  $f$  is a continuous map of  $[A, B]$  into  $V$ ,  $-\infty < A < B < +\infty$ , and  $\varepsilon$  is positive, there exists  $\sum_0^n c_k x^k$  with coefficients  $c_k$  in  $V$  such that  $\|f(x) - \sum_0^n c_k x^k\| < \varepsilon$ ,  $A \leq x \leq B$ .

*Dirichlet problem for an annulus.* Let  $A = \{r < |z| < s\}$ ,  $0 < r < s < +\infty$ , and suppose that  $f$  is a real-valued function continuous on  $C(0; r)$  and  $g$  is a real-valued function continuous on  $C(0; s)$ . We seek  $u$  real-valued continuous on  $\bar{A}$ , harmonic in  $A$ , and satisfying the boundary conditions  $u|_{C(0; r)} = f$ ,  $u|_{C(0; s)} = g$  (Dirichlet problem for annulus). Applying the above approximation theorem, we see that given  $\varepsilon$  positive there exist a sum  $\sum_{-n}^n a_k (z/r)^k$  and a sum  $\sum_{-n}^n b_k (z/s)^k$  such that

$$\left| f(z) - \sum_{-n}^n a_k \left(\frac{z}{r}\right)^k \right| < \varepsilon, \quad |z| = r,$$

$$\left| g(z) - \sum_{-n}^n b_k \left(\frac{z}{s}\right)^k \right| < \varepsilon, \quad |z| = s,$$



and  $a_{-k} = \bar{a}_k$ ,  $b_{-k} = \bar{b}_k$ ,  $k = 0, \dots, n$ . We seek to determine coefficients  $\alpha$ ,  $\beta$ ,  $\gamma_k$  such that

$$U(z) = \alpha + \beta \log |z| + \sum_1^n (\gamma_k z^k + \bar{\gamma}_k \bar{z}^k + \gamma_{-k} z^{-k} + \bar{\gamma}_{-k} \bar{z}^{-k})$$

reduces to  $\sum_{-n}^n a_k (z/r)^k$  on  $C(0; r)$  and to  $\sum_{-n}^n b_k (z/s)^k$  on  $C(0; s)$ . It suffices that the following conditions be fulfilled:

$$\begin{aligned} \alpha + \beta \log r &= a_0, & \alpha + \beta \log s &= b_0; \\ \gamma_k + \bar{\gamma}_{-k} r^{-2k} &= a_k r^{-k}, \\ \gamma_k + \bar{\gamma}_{-k} s^{-2k} &= b_k s^{-k}, & k &= 1, \dots, n. \end{aligned}$$

It is clear that there exist coefficients satisfying these conditions. We take a sequence of  $U$  such that the corresponding sequence of restrictions to  $C(0; r)$  (resp.  $C(0; s)$ ) tends uniformly to  $f$  (resp.  $g$ ). By the boundary maximum principle for harmonic functions, Ex. 8.5, Ch. IV, we see that the sequence is uniform Cauchy on  $\bar{A}$ . It tends to the solution of the proposed Dirichlet problem. Related approximation methods have been used by Julia (*Atti del Congr. Intern. Mat., Bologna, 1928, 95–101*).

## EXERCISE

2.2. Determine the coefficients  $a_k$ ,  $b$  in the representation

$$b \log |z| + \sum_{-\infty}^{+\infty} (a_k z^k + \bar{a}_k \bar{z}^k), \quad r < |z| < s,$$

of  $h|_A$ , where  $h$  is the solution of the above Dirichlet problem, in terms of  $f$  and  $g$  [cf. (3.3), Ch. XI]. Here  $a_0$  is taken as real. Hint: Consider the complex Fourier coefficients of  $h(\rho e^{i\theta})$ ,  $r < \rho < s$ .

## 3. PRINCIPLES OF SCHWARZ AND CARLEMAN

Suppose that  $f$  is a complex-valued function continuous on

$$A = \Delta(0; 1) \cap \{\operatorname{Im} z \geq 0\}$$

in the interior of which it is analytic and that  $f(x)$  is real,  $-1 < x < 1$ . It is easy to show that there exists a function  $g$  analytic on  $\Delta(0; 1)$  whose restriction to  $A$  is  $f$ .† If such a  $g$  exists, then  $\overline{g(\bar{z})}$  with domain  $\Delta(0; 1)$  is also analytic on  $\Delta(0; 1)$  and agrees with  $g$  on  $] -1, 1[$ . Hence  $g(z) = \overline{g(\bar{z})}$ . This suggests that

† It is clear that there is at most one such  $g$ .

we propose  $g$  defined by  $g|_A = f$ ,  $g(z) = \overline{f(\bar{z})}$  for  $z \in \Delta(0; 1) \cap \{\operatorname{Im} z < 0\}$ . The function so defined is continuous on  $\Delta(0; 1)$  and analytic in  $\Delta(0; 1) - \mathbf{R}$ . Hence by the second theorem of §7, Ch. V, it is analytic on  $\Delta(0; 1)$ .

The Schwarz reflexion principle is a parallel result for harmonic functions. In fact, it even allows us to draw the above conclusion for  $f$  satisfying the less stringent:  $f$  analytic on  $\Delta(0; 1) \cap \{\operatorname{Im} z > 0\}$ ;  $\lim_a \operatorname{Im} f = 0$ ,  $-1 < a < 1$ . The Schwarz reflexion principle states:

*Let  $u$  be continuous on  $A = \Delta(0; 1) \cap \{\operatorname{Im} z \geq 0\}$ , be harmonic in the interior of  $A$ , and satisfy  $u(x) = 0$ ,  $-1 < x < 1$ . Then there exists a unique function  $h$  harmonic on  $\Delta(0; 1)$  satisfying  $u = h|_A$ .*

The uniqueness is immediate (cf. Ex. 3.3, Ch. XI). To establish existence, let  $0 < r < 1$  and let

$$v(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} U(re^{i\theta}) k(\theta, z/r) d\theta, \quad |z| < r,$$

where  $U(re^{i\theta}) = u(re^{i\theta})$ ,  $0 \leq \theta \leq \pi$ , and  $U(re^{i\theta}) = -u(re^{-i\theta})$ ,  $-\pi \leq \theta < 0$ . Now  $v(\bar{z}) = -v(z)$  as we see on examining the Poisson integral and hence  $v(z) = u(z)$ ,  $|z| \leq r$ ,  $\operatorname{Im} z \geq 0$ . We see that the union of the  $v$  is the desired  $h$ .

In the function-theoretic situation for which it was indicated that the Schwarz reflexion principle was applicable, we see that there exists  $h$  harmonic on  $\Delta(0; 1)$  such that  $h(z) = \operatorname{Im} f(z)$  on the domain of  $f$ . Let  $g$  be analytic on  $\Delta(0; 1)$  and satisfy  $h = \operatorname{Im} g$ . Then for some real constant  $c$  the function  $g + c$  has restriction  $f$  on the domain of  $f$ .

Carleman's extension principle states:

*Let  $f$  be analytic on  $A = \Delta(0; 1) \cap \{\operatorname{Im} z > 0\}$  and let  $g$  be analytic on  $B = \Delta(0; 1) \cap \{\operatorname{Im} z < 0\}$ . Suppose that*

$$\lim_a [f(z) - g(\bar{z})] = 0, \quad (3.1)$$

*$-1 < a < 1$ . Then there exists a unique function  $h$  analytic on  $\Delta(0; 1)$  such that  $f = h|_A$  and  $g = h|_B$ .*

PROOF:  $i[f(z) - \overline{g(\bar{z})}]$  and  $f(z) + \overline{g(\bar{z})}$  are analytic on  $A$  and the imaginary part of each has limit 0 at each point of  $] -1, 1[$ . We conclude that

$$f(z) - \overline{g(\bar{z})} = F(z), \quad f(z) + \overline{g(\bar{z})} = G(z), \quad z \in A,$$

where  $F$  and  $G$  are analytic on  $\Delta(0; 1)$ . Setting

$$h(z) = \frac{F(z) + G(z)}{2},$$

$$l(z) = \frac{\overline{G(\bar{z})} - \overline{F(\bar{z})}}{2},$$

$|z| < 1$ , we have  $f = h|_A$ ,  $g = l|_B$ . By (3.1) we have  $h(a) = l(a)$ ,  $-1 < a < 1$ . Hence  $h = l$ . The uniqueness assertion is obvious. Carleman's principle follows.

It should be remarked that Carleman also gave more refined versions of his principle (cf. [25]).

## EXERCISES

**3.1.** Let  $f: \mathbf{R} \rightarrow \mathbf{C}$ . Show that if  $(g, h)$  satisfies  $g$  analytic on  $\{\operatorname{Im} z > 0\}$ ,  $h$  analytic on  $\{\operatorname{Im} z < 0\}$ ,  $f(a) = \lim_a [g(z) + h(\bar{z})]$  for all  $a \in \mathbf{R}$ , then  $g$  and  $h$  are determined additively up to the same entire function; that is, if  $(g_1, h_1)$  also satisfies the stated conditions with  $g_1$  replacing  $g$  and  $h_1$  replacing  $h$ , there exists  $\varphi$  entire such that  $g_1(z) = g(z) + \varphi(z)$ ,  $\operatorname{Im} z > 0$ , and  $h_1(z) = h(z) - \varphi(z)$ ,  $\operatorname{Im} z < 0$ . If  $g$  and  $h$  are considered which satisfy

$$\lim_{\infty} z^{-1} g(z) = \lim_{\infty} z^{-1} h(z) = 0,$$

then they are determined additively up to a constant.

**3.2.** Show that if  $f: \mathbf{R} \rightarrow \mathbf{C}$  is continuous on  $\mathbf{R}$  and bounded, then  $f$  admits a representation of the type given in the second sentence of the previous exercise.

## 4. SUBHARMONIC FUNCTIONS

The theory of subharmonic functions may be approached at different levels of generality. We shall be content to proceed under a hypothesis of continuity quite adequate for the applications to be considered.

*Some preliminaries.* Given  $X \neq \emptyset$  suppose that  $f_k: X \rightarrow \hat{\mathbf{R}}$ ,  $k = 1, \dots, n$  (where  $n$  is a positive integer). We define  $\max\{f_1, \dots, f_n\}$ , the *maximum* of  $f_1, \dots, f_n$ , as the function with domain  $X$  assigning to  $x$  the value  $\max\{f_1(x), \dots, f_n(x)\}$ . In spite of the similarity of notation the sense will be clear from context. Given  $f: X \rightarrow \hat{\mathbf{R}}$  and  $g: X \rightarrow \mathbf{R}$  we define  $f + g$  and  $f - g$ , as would be expected, by taking  $f + g$  as the function  $f(x) + g(x)$  with domain  $X$  and  $f - g$  as  $f + (-g)$ .† We shall also want a simple definition for integrals of functions which are continuous maps of a segment  $[a, b]$  into  $\mathbf{R} \cup \{-\infty\}$ . For such a function  $f$  by

$$\int_a^b f dx$$

we shall understand the limit of the sequence the  $n$ th term of which is the

† It is to be observed that our earlier conventions did not cover the situation at hand.

integral on  $[a, b]$  of the maximum of  $f$  and the constant  $-n$ ,  $n \in \mathbb{N}$ . When  $f$  is finite-valued the integral as just given coincides with the definition given earlier.

We consider  $O$  open  $\subset \mathbb{C}$ . We say that a continuous map  $u$  of  $O$  into  $\mathbb{R} \cup \{-\infty\}$  is *subharmonic* on  $O$  (more briefly, *subharmonic*) provided that for each  $a \in O$  we have

$$u(a) \leq \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta \quad (4.1)$$

for sufficiently small positive  $r$  (mean-value property). Our prior experience with the Gauss mean-value property of harmonic functions should make it apparent that the apparatus of the theory of harmonic functions can be brought to bear on the theory of subharmonic functions. Since the modulus of a function analytic on  $O$  satisfies (4.1) as does the norm of a vector-valued function analytic on  $O$ , we may anticipate the utility of the theory of subharmonic functions for complex function theory.

We start our study of subharmonic functions with the important *principle of the harmonic majorant*:

*Let  $u$  be subharmonic on  $O$  and let  $h$  be harmonic on  $O$  and satisfy:*

$$\limsup_w (u - h) \leq 0 \quad (4.2)$$

*for each  $w \in \text{fr } O$  where  $\text{fr}$  is taken in the sense of  $\hat{\mathbb{C}}$ . Then*

$$u(z) \leq h(z), \quad z \in O.$$

Here “ $\limsup$ ” is construed in the sense of the relative topology of  $O \cup \{w\}$ .

We note that this result subsumes the boundary maximum principles for harmonic functions and analytic functions. Thus if  $f$  is a continuous map of  $\bar{O}$  (in the sense of  $\hat{\mathbb{C}}$ ) into  $\mathbb{C}$  which is analytic in  $O$ , then  $\max_{\bar{O}} |f|$  is the maximum of  $(|f|) | \text{fr } O$ . To see that this assertion follows from the principle of the harmonic majorant it suffices to take  $u = (|f|) | O$  and  $h$  the constant whose value is the maximum of  $(|f|) | \text{fr } O$ . The same observation holds for harmonic functions.

Let us now prove the principle of the harmonic majorant. Let  $v = u - h$ . If the assertion were false, there would exist  $a \in O$  such that  $v(a) = \max_O v > 0$ . Let  $\Omega$  denote the component of  $O$  containing  $a$ . Since  $v$  is subharmonic, as we readily verify (considering separately the cases  $v(z) = -\infty$  and  $v(z) > -\infty$ ), we see that as a consequence of the continuity of  $v$  and the mean-value property both  $\{z \in \Omega, v(z) < v(a)\}$  and  $\{z \in \Omega, v(z) = v(a)\}$  are open. Hence  $v(z) = v(a)$ ,  $z \in \Omega$ . But  $\text{fr } \Omega$  is not empty and is contained in  $\text{fr } O$ . Applying (4.2) at  $w \in \text{fr } \Omega$  we obtain a contradiction. The principle follows.

## EXERCISE

**4.1.** Show that if  $u$  and  $v$  are subharmonic on  $O$ , then so are  $u + v$  and  $\max\{u, v\}$ . [Here addition is pointwise;  $u + v$  is the function  $u(z) + v(z)$  with domain  $O$ .]

*Poisson modification.* We continue with  $u$  subharmonic on  $O$  and consider  $\Delta(a; r) \subset O$ ,  $0 < r < +\infty$ . We define  $v$  with domain  $O$  by

$$\begin{aligned} v(z) &= u(z), & z \in O - \Delta(a; r), \\ &= \lim_{n \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} u_n(a + re^{i\theta}) k\left(\theta, \frac{z-a}{r}\right) d\theta, & |z-a| < r, \end{aligned}$$

where  $u_n$  is the maximum of  $u$  and the constant  $-n$ . Using the principle of the harmonic majorant and the Harnack convergence principle (Ex. 1.5, this chapter) we see that  $v(z) \geq u(z)$ ,  $|z-a| < r$ , and that  $v$  is harmonic in  $\Delta(a; r)$  when  $u$  takes a value  $> -\infty$  at some point of  $\Delta(a; r)$ , and otherwise  $v$  takes the value  $-\infty$  at each point of  $\Delta(a; r)$ . We now show that  $v$  is *subharmonic*. It suffices to test the continuity of  $v$  at points of  $C(a; r)$ , for the continuity and the mean-value property at other points are assured and the mean-value property of  $v$  at points of  $C(a; r)$  will follow from the fact that  $v(z) \geq u(z)$ ,  $z \in O$ , with equality holding on  $C(a; r)$ . Now

$$u(z) \leq \liminf_z v \leq \limsup_z v \leq u_n(z), \quad z \in C(a; r),$$

and the asserted continuity follows by the arbitrariness of  $n$ .

We term  $v$  the *Poisson modification of  $u$  associated with  $\Delta(a; r)$* .

## EXERCISE

**4.2.** Show that if  $\text{int}\{u(z) = -\infty\} \neq \emptyset$ , where  $u$  is subharmonic on a region  $A$ , then  $u$  is the constant  $-\infty$ . Hint: Standard connectedness argument.

*Perron family.* Let  $A$  be a region in  $\mathbb{C}$ . A family  $\Phi$  of functions subharmonic on  $A$  is said to be a *Perron family* provided that the following two conditions are fulfilled: (1) If  $u, v \in \Phi$ , then  $\max\{u, v\} \in \Phi$ . (2) If  $a \in A$ , then for each  $r$ ,  $0 < r < +\infty$ , such that  $\Delta(a; r) \subset A$ , for each  $u \in \Phi$  the Poisson modification of  $u$  associated with  $\Delta(a; r)$  belongs to  $\Phi$ . Let

$$M(z) = \sup\{u(z) : u \in \Phi\}, \quad z \in A.$$

We prove the following theorem concerning  $\Phi$ .

**Theorem 4.1:** *One of the following holds:  $M$  is the constant  $-\infty$ ,  $M$  is the constant  $+\infty$ ,  $M$  is harmonic on  $A$ .*

**PROOF:** Let  $a \in A$ . If  $M(a) = -\infty$ , then either  $\Phi = \emptyset$  or  $\Phi$  contains only the constant  $-\infty$ . For if  $u \in \Phi$  and  $u$  is not the constant  $-\infty$ , some Poisson modification of  $u$  takes a finite value at  $a$  so that  $M(a) > -\infty$ . We see that if  $\overline{M(a)} = -\infty$ , then  $M$  is the constant  $-\infty$ . Suppose that  $M(a) = +\infty$ . Let  $\Delta(a; r) \subset A$ ,  $0 < r < +\infty$ . Let  $(u_n): \mathbf{N} \rightarrow \Phi$  satisfy  $\lim_{n \rightarrow \infty} u_n(a) = +\infty$ , all  $u_n(a)$  being  $> -\infty$ . Let  $v_n = \max\{u_0, \dots, u_n\}$  and let  $w_n$  be the Poisson modification of  $v_n$  associated with  $\Delta(a; r)$ . Then  $w_n \in \Phi$ ,  $w_n$  is harmonic in  $\Delta$ ,  $w_n(z) \leq w_{n+1}(z)$  for  $z \in A$ ,  $n \in \mathbf{N}$ , and  $\lim_{n \rightarrow \infty} w_n(a) = +\infty$ . Hence by the Harnack convergence principle (Ex. 1.5, this chapter)  $M(z) = +\infty$ ,  $z \in \Delta(a; r)$ . Consequently  $\{M(z) = +\infty\}$  is open. If  $M(a) \in \mathbf{R}$ , given  $b \in \Delta(a; r)$ , we take  $(u_n): \mathbf{N} \rightarrow \Phi$  satisfying  $\lim_{n \rightarrow \infty} u_n(b) = M(b)$  and construct  $v_n$  and  $w_n$  as before. Then  $\lim_{n \rightarrow \infty} w_n(b) = M(b)$ , and consequently, since  $w_n(a) \leq M(a)$ , by the Harnack convergence principle  $M(b) < +\infty$ . Hence  $\{M(z) \in \mathbf{R}\}$  is open. We conclude that at least one of the following is true:  $M$  is the constant  $-\infty$ ,  $M$  is the constant  $+\infty$ ,  $M$  takes values in  $\mathbf{R}$  only. Of course, the connectedness of  $A$  is essential to the argument.

Suppose now that  $M$  takes values in  $\mathbf{R}$  only. We continue with  $\Delta(a; r)$  as above,  $b \in \Delta(a; r)$ . Let  $(u_n), (U_n): \mathbf{N} \rightarrow \Phi$  be such that  $\lim_{n \rightarrow \infty} u_n(a) = M(a)$ ,  $\lim_{n \rightarrow \infty} U_n(b) = M(b)$ . Let  $v_n = \max\{u_0, \dots, u_n\}$ , let

$$V_n = \max\{u_0, \dots, u_n, U_0, \dots, U_n\},$$

let  $w_n$  denote the Poisson modification of  $v_n$  associated with  $\Delta(a; r)$ , and let  $W_n$  denote the Poisson modification of  $V_n$  associated with  $\Delta(a; r)$ . Now  $(w_n)$  tends pointwise on  $A$  to a function  $w$  and  $(W_n)$  tends pointwise on  $A$  to a function  $W$ . The functions  $w$  and  $W$  are harmonic in  $\Delta(a; r)$  and satisfy  $w(z) \leq W(z)$ ,  $z \in \Delta(a; r)$  and  $w(a) = W(a) = M(a)$ . Hence  $w(z) = W(z)$ ,  $z \in \Delta(a; r)$ , by the maximum principle for harmonic functions. Since  $M(b) = W(b)$ ,  $M(b) = w(b)$ , and we see that  $M$  is harmonic in  $\Delta(a; r)$ . By the arbitrariness of  $a$ ,  $M$  is harmonic on  $A$ . Theorem 4.1 follows.

## EXERCISE

**4.3.** Show that Theorem 4.1 remains valid when “for each  $r$ ” is replaced by “for some  $r$ ” in condition (2) for a Perron family. The more restrictive definition is however useful.

*An application.* We shall apply Theorem 4.1 later appropriately recast for Riemann surfaces to solve a simple Dirichlet problem incidental to the study

of the problem of Riemann and Klein. For the present we show how the theorem just proved and the solution of the Dirichlet problem for an annulus permit us to treat the Dirichlet problem for a bounded region  $A$  in  $\mathbb{C}$  for which

$$\text{fr } A = \bigcup_{1 \leq k \leq n} C(a_k; r_k)$$

where

$$C(a_j; r_j) \cap C(a_k; r_k) = \emptyset, \quad j \neq k,$$

and all  $r_k$  are positive.

Suppose that  $f_k$  is a continuous map of  $C(a_k; r_k)$  into  $\mathbb{R}$ ,  $k = 1, \dots, n$ . We seek  $h$  continuous on  $\bar{A}$ , harmonic in  $A$  and satisfying  $h|C(a_k; r_k) = f_k$ ,  $k = 1, \dots, n$  (Dirichlet problem for  $A$ ). To that end we introduce for each  $k$  an annulus  $A_k \subset A$  where  $\text{fr } A_k$  consists of  $C(a_k; r_k)$  and  $C(a_k; s_k)$  for some positive  $s_k$  and  $\bar{A}_j \cap \bar{A}_k = \emptyset$ ,  $j \neq k$ . We let  $m$  denote a real number less than all the values taken by the  $f_k$  and the  $(-f_k)$ . Let  $v$  be defined on  $\bar{A}$  by the requirements that  $v| \bar{A}_k$  be the solution of the Dirichlet problem for  $A_k$  reducing to  $f_k$  on  $C(a_k; r_k)$  and to the constant  $m$  on  $C(a_k; s_k)$ ,  $k = 1, \dots, n$ , and that  $v$  take the value  $m$  at each point of  $A - \bigcup_{1 \leq k \leq n} \bar{A}_k$ . Let  $w$  be similarly defined with  $(-f_k)$  replacing  $f_k$ . Both  $v|A$  and  $w|A$  are subharmonic. Applying the principle of the harmonic majorant to  $(v + w)|A$  with comparison function the constant 0 we see that  $v(z) \leq -w(z)$ ,  $z \in A$ . Now let  $\Phi$  denote the family of functions  $u$  subharmonic on  $A$  satisfying:

$$v(z) \leq u(z) \leq -w(z), \quad z \in A. \quad (4.3)$$

Clearly  $v|A \in \Phi$ . Now  $\Phi$  is a Perron family. Indeed, it is immediate that if  $u_1, u_2 \in \Phi$ , then  $\max\{u_1, u_2\} \in \Phi$ , as we see by reference to (4.3). If  $u \in \Phi$  and  $U$  is a Poisson modification of  $u$ , then (4.3) with  $U$  replacing  $u$  holds. The inequality on the left is obvious and the principle of the harmonic majorant applied to  $U + w|A$  with comparison function the constant 0 shows that the inequality on the right holds. By Theorem 4.1 the function  $M$  is harmonic on  $A$ . Since  $M$  replacing  $u$  satisfies (4.3), we see that  $M \cup \bigcup_{1 \leq k \leq n} f_k$  is the desired function  $h$ .

## EXERCISES

These two exercises give further experience with some elementary aspects of the notion of a subharmonic function.

4.4. Suppose that  $u \in C''[\Delta(0; r)]$  and is real-valued. Let

$$\mu(s) = \frac{1}{2\pi} \int_0^{2\pi} u(se^{i\theta}) d\theta,$$

$0 \leq s < r$ . Show that  $\mu'$  is continuous on  $[0, r[$ . Show by integrating  $\Delta u(se^{i\theta})$  with respect to  $\theta$  on  $[0, 2\pi]$  and employing standard theorems of the differential calculus that if  $\Delta u(z) = 4u_{z\bar{z}}(z) \geq 0$ ,  $|z| < r$ , then  $\mu'(s) \geq 0$ ,  $0 \leq s < r$ , and if  $\Delta u(z) > 0$ ,  $|z| < r$ , then  $\mu'(s) > 0$ ,  $0 < s < r$ . Conclude that if now  $u \in C''(O)$  and is real-valued, then  $u$  is subharmonic if and only if  $\Delta u(z) \geq 0$ ,  $z \in O$ . Here  $O$  is an open subset of  $\mathbb{C}$ . The  $C''$  notion in the present setting is to be appropriately formulated.

**4.5.** Let  $-\infty \leq a < b \leq +\infty$  and let  $f$  be convex on  $]a, b[$  and satisfy  $f(x) \leq f(y)$ ,  $a < x < y < b$ . Let  $u$  be subharmonic on  $O$  (as above) and map  $O$  into  $]a, b[$ . Show that  $f \circ u$  is subharmonic.

Hint: Use is to be made of the following elementary theorem: If  $g$  is a continuous map of  $[A, B]$  into  $[a, b]$  where  $a, b, A, B$  are real and  $a < b$ ,  $A < B$ , and  $f$  is continuous and convex on  $[a, b]$ , then

$$f \left[ (B - A)^{-1} \int_A^B g \, dx \right] \leq (B - A)^{-1} \int_A^B f \circ g \, dx.$$

This inequality may be deduced by using approximating sums for the integral of  $g$  and noting the barycenter property of the convex set

$$E = \{(x, y): a \leq x \leq b, y \geq f(x)\},$$

namely, if  $(x_k, y_k) \in E$ ,  $\mu_k \geq 0$ ,  $k = 1, \dots, n$ , and  $\sum_1^n \mu_k = 1$ , then

$$\left( \sum_1^n \mu_k x_k, \sum_1^n \mu_k y_k \right) \in E.$$